

JavaScript DOM Events — Answers 1–15

Complete HTML and JavaScript solutions for all requested tasks.

Task 1

Create an HTML file with a button labeled 'Add to Playlist' and use onclick to display an alert saying 'Song added!' when clicked.

```
<!DOCTYPE html>
<html>
<body>
  <button onclick="alert('Song added!')">Add to Playlist</button>
</body>
</html>
```

Task 2

Create an HTML file with a button labeled 'Add to Playlist' and use onclick to display an alert saying 'Song added!' when clicked.

```
<!DOCTYPE html>
<html>
<body>
  <button onclick="alert('Song added!')">Add to Playlist</button>
</body>
</html>
```

Task 3

Create an HTML page with an 'Add Song' button that instantly adds a new card below showing a random song title from a favorite playlist.

```
<!DOCTYPE html>
<html>
<head>
  <title>Add Song</title>
  <style>
    .card {
      width: 250px;
      padding: 15px;
      margin-top: 10px;
      border: 1px solid #ccc;
      border-radius: 8px;
    }
  </style>
</head>
<body>
  <button onclick="addSong()">Add Song</button>
  <div id="songs"></div>

  <script>
    const playlist = [
      "Blinding Lights",
      "Perfect",
      "Shape of You",
      "Believer",
      "Levitating"
    ];

    function addSong() {
      const randomSong =
        playlist[Math.floor(Math.random() * playlist.length)];

      const card = document.createElement("div");
      card.className = "card";
      card.textContent = randomSong;

      document.getElementById("songs").appendChild(card);
    }
  </script>
</body>
</html>
```

Task 4

Build a mini form with an input for 'Username' and use onchange to log the entered username to the console every time the input changes.

```
<!DOCTYPE html>
<html>
<body>
  <label>Username:</label>
  <input type="text" onchange="logUsername(this.value)">

  <script>
    function logUsername(username) {
      console.log("Username:", username);
    }
  </script>
</body>
</html>
```

Task 5

Build a mini form with an input for 'Username' and use onchange to log the entered username to the console every time the input changes.

```
<!DOCTYPE html>
<html>
<body>
  <label>Username:</label>
  <input type="text" onchange="logUsername(this.value)">

  <script>
    function logUsername(username) {
      console.log("Username:", username);
    }
  </script>
</body>
</html>
```

Task 6

Build a simple Zomato-style food order form with a cuisine dropdown and quantity input. Use onchange on the dropdown to display which cuisine was selected.

```
<!DOCTYPE html>
<html>
<head>
  <title>Food Order</title>
</head>
<body>
  <h2>Food Order</h2>

  <label>Cuisine:</label>
  <select id="cuisine" onchange="showCuisine(this.value)">
    <option value="">Select cuisine</option>
    <option value="Indian">Indian</option>
    <option value="Chinese">Chinese</option>
    <option value="Italian">Italian</option>
    <option value="Mexican">Mexican</option>
  </select>

  <br><br>

  <label>Quantity:</label>
  <input type="number" min="1" value="1">

  <p id="message"></p>

  <script>
    function showCuisine(cuisine) {
      document.getElementById("message").textContent =
        cuisine ? "You selected: " + cuisine : "";
    }
  </script>
</body>
</html>
```

Task 7

Create an Instagram-style Like button using `addEventListener` to increment and display the like count on every click.

```
<!DOCTYPE html>
<html>
<body>
  <button id="likeBtn">♥■ Like: 0</button>

  <script>
    let likes = 0;
    const likeBtn = document.querySelector("#likeBtn");

    likeBtn.addEventListener("click", function () {
      likes++;
      likeBtn.textContent = "♥■ Like: " + likes;
    });
  </script>
</body>
</html>
```

Task 8

Create an Instagram-style Like button using `addEventListener` to increment and display the like count on every click.

```
<!DOCTYPE html>
<html>
<body>
  <button id="likeBtn">♥■ Like: 0</button>

  <script>
    let likes = 0;
    const likeBtn = document.querySelector("#likeBtn");

    likeBtn.addEventListener("click", function () {
      likes++;
      likeBtn.textContent = "♥■ Like: " + likes;
    });
  </script>
</body>
</html>
```

Task 9

Write a function `addToCart` that adds a product name to a visible cart section when an 'Add to Cart' button is clicked. Attach the click event using `addEventListener`.

```
<!DOCTYPE html>
<html>
<body>
  <h2>Product</h2>
  <p>Product: Wireless Headphones</p>
  <button id="cartBtn">Add to Cart</button>

  <h3>Cart</h3>
  <ul id="cart"></ul>

  <script>
    function addToCart(productName) {
      const item = document.createElement("li");
      item.textContent = productName;
      document.querySelector("#cart").appendChild(item);
    }

    document.querySelector("#cartBtn").addEventListener("click", function () {
      addToCart("Wireless Headphones");
    });
  </script>
</body>
</html>
```

Task 10

Display 3 hardcoded movies with a Remove button next to each. Use addEventListener to remove the movie card from the DOM.

```
<!DOCTYPE html>
<html>
<head>
  <style>
    .movie {
      padding: 10px;
      margin: 8px 0;
      border: 1px solid #ccc;
      width: 300px;
    }
  </style>
</head>
<body>
  <div id="movies"></div>

  <script>
    const movieNames = ["Inception", "Interstellar", "The Dark Knight"];
    const movies = document.querySelector("#movies");

    movieNames.forEach(function (name) {
      const card = document.createElement("div");
      card.className = "movie";

      const title = document.createElement("span");
      title.textContent = name + " ";

      const removeBtn = document.createElement("button");
      removeBtn.textContent = "Remove";

      removeBtn.addEventListener("click", function () {
        card.remove();
      });

      card.append(title, removeBtn);
      movies.appendChild(card);
    });
  </script>
</body>
</html>
```

Task 11

Display 3 hardcoded movies with a Remove button next to each. Use addEventListener to remove the movie card from the DOM.

```
<!DOCTYPE html>
<html>
<head>
  <style>
    .movie {
      padding: 10px;
      margin: 8px 0;
      border: 1px solid #ccc;
      width: 300px;
    }
  </style>
</head>
<body>
  <div id="movies"></div>

  <script>
    const movieNames = ["Inception", "Interstellar", "The Dark Knight"];
    const movies = document.querySelector("#movies");

    movieNames.forEach(function (name) {
      const card = document.createElement("div");
      card.className = "movie";

      const title = document.createElement("span");
      title.textContent = name + " ";

      const removeBtn = document.createElement("button");
      removeBtn.textContent = "Remove";

      removeBtn.addEventListener("click", function () {
        card.remove();
      });
    });
  </script>
</body>
</html>
```

```

        card.append(title, removeBtn);
        movies.appendChild(card);
    });
</script>
</body>
</html>

```

Task 12

Display the X and Y coordinates of the mouse when the user clicks anywhere inside a div, using `event.clientX` and `event.clientY`.

```

<!DOCTYPE html>
<html>
<head>
<style>
    #area {
        width: 500px;
        height: 250px;
        border: 2px solid black;
    }
</style>
</head>
<body>
<div id="area"></div>
<p id="coordinates">Click inside the box.</p>

<script>
    document.querySelector("#area").addEventListener("click", function (event) {
        document.querySelector("#coordinates").textContent =
            "X: " + event.clientX + ", Y: " + event.clientY;
    });
</script>
</body>
</html>

```

Task 13

Log the type and coordinates (`clientX`, `clientY`) of any mouse event when the user interacts with a Flipkart-style 'Add to Cart' button.

```

<!DOCTYPE html>
<html>
<body>
<button id="addCart">Add to Cart</button>

<script>
    const button = document.querySelector("#addCart");

    function logMouseEvent(event) {
        console.log(
            "Type:", event.type,
            "X:", event.clientX,
            "Y:", event.clientY
        );
    }

    button.addEventListener("click", logMouseEvent);
    button.addEventListener("mouseover", logMouseEvent);
    button.addEventListener("mousedown", logMouseEvent);
</script>
</body>
</html>

```

Task 14

Log the type and coordinates (`clientX`, `clientY`) of any mouse event when the user interacts with a Flipkart-style 'Add to Cart' button.

```

<!DOCTYPE html>
<html>
<body>
<button id="addCart">Add to Cart</button>

<script>
    const button = document.querySelector("#addCart");

```

```
function logMouseEvent(event) {
  console.log(
    "Type:", event.type,
    "X:", event.clientX,
    "Y:", event.clientY
  );
}

button.addEventListener("click", logMouseEvent);
button.addEventListener("mouseover", logMouseEvent);
button.addEventListener("mousedown", logMouseEvent);
</script>
</body>
</html>
```

Task 15

Briefly explain in 2–3 sentences what event flow is in the DOM, and give one example of how event bubbling could affect a shopping cart app like Flipkart.

Event flow is the process by which a DOM event travels through elements, commonly through the capturing phase, the target phase, and the bubbling phase. In event bubbling, an event triggered on a child element can move upward to its parent elements; for example, if an Add to Cart button is inside a product card with a click listener, clicking the button can also trigger the card's click handler unless propagation is controlled with `event.stopPropagation()`.